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TEACHING MANUAL 1

STD 8 unit 3 The Trail of Science **Date:**

Name of the teacher:

Lesson: Talking of Science

prds. 14

Theme:Life and vision of a scientist

Sub themes:Curiosity and scientific temper,innovation and social responsibility etc

Learning outcomes:

- **Recognize the qualities, struggles, and achievements of a scientist.**
- **Identify the importance of curiosity, perseverance, and dedication in scientific research.**
- **Enhance vocabulary related to science, research, and innovation.**
- **Improve speaking and listening skills through role-play**
- **Cultivate a spirit of inquiry and critical thinking.**
- **Frame interview questions and conduct a mock interview**
- **communicate effectively in the target language.**

Resources:

videos, pictures slides, CB

Process	Assessment
Introductory interaction **Teacher conducts a quiz related to scientists and inventions .Shows a video about Ajith Parameswaran** **Initiates a discussion** ● **Who is your favourite scientist? Why do you admire him/her?** ● **Can you name a few scientific discoveries or inventions that changed the world?** ● **What qualities do you think are important for a good scientist?** ● **Have you noticed anything special in the attitude of Dr Ajith towards science?** **Teacher asks the slow pace learners to say what they feel special about the man** **Elicits responses**	

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Here is an interesting interview with Dr Ajith

Process reading

Teacher proceeds to the text. They are asked to read the first two questions

They keep track of the reading .Teacher supports the reading process by showing some images to convey the meaning of difficult towards. initiates a discussion.

- According to Dr. Ajith, what is a good description of the work scientists do?
- How does Dr. Ajith compare scientists with farmers, vendors, and engineers?
- What role does asking questions play in scientific exploration?
- Why do you think Dr. Ajith asks to be called simply "Ajith" instead of "Dr. Parameswaran"?

Elicits responses.

Discourse

Asks about the features of a good interview and to prepare a few relevant questions for an interview with a prominent person in the locality.

Teacher includes slow pace learners and asks them to take part in it ,They attempt the discourse, group refinement,presentation Editing. Teacher records the presentations and suggests modifications.

Teacher proceeds to the next section. They are asked to read next three questions

They keep track of their reading. teacher asks some global questions to ensure reading. learners sit in groups and share their findings. they refer to the glossary. Teacher supports the reading process in groups. Teacher shows some slides to convey the meanings of words

initiates the discussion

- Curiosity is at the heart of science.' Do you agree?

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- **How does observation help us understand natural phenomena?**
- **What are the unsolved mysteries you would like to solve, with the help of science?**

Elicits responses
Teacher includes the slow pace learners by asking them simple questions
Discourse
What is the purpose of a notice? Which are the different types of notices? What are the key features of a notice? Discuss.
As the Secretary of the Science Club of your school, you plan to conduct a science exhibition. Prepare a notice giving details of the programme. They attempt the discourse, group refinement, presentation Editing. Teacher records the presentations and suggests modifications.
Assignment
Human-animal conflicts and extreme weather events such as flash floods and landslides are on the increase today. Try to list some of the probable causes of these events. List some mysteries of Nature that remain unsolved. Write a short note on any one of these mysteries.
Teacher proceeds with the next section .They are asked to read the last part of the interview They keep track of their reading. Teacher asks some global questions to ensure reading. Teacher gives some contexts and some images to get the meanings. Initiates a discussion

- **What are the essential qualities of a good scientist, according to Ajith?**
- **Do you agree with Ajith's statement that *"anyone can become a scientist"*? Why or why not?**
- **What message does Ajith give to youngsters about following their interests?**
- **How does Ajith connect passion, hard work, and success in his advice to students?**

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• **What do you understand by the phrase *"standing on the shoulders of giants"* in the context of science?** **Elicits responses.** ***Discourse*** **let's attempt a 'Just A Minute' (JAM) speech about scientific inventions that changed the world, using the following indicators.** **•The name of the inventor(s)** **•Impact of the invention/discovery** **•Positive and negative effects** **Assignment** **Activity 6** **Cracking the Code: The Wonder** ***We use a number of gadgets in our daily life. Pick any gadget of your choice and describe its features.***	

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TEACHING MANUAL 2

STD 8 The Trail of Science **Date:**

Name of the teacher:

Lesson: Engine Trouble

prds. 14

Theme:Irony of life

Sub themes:The burden of ownership,helplessness of the common man ,role of luck and chance etc

Learning outcomes:

- **Analyse how irony and humour are used to highlight deeper truths of human life.**
- **Appreciate Narayan's storytelling style that mixes humour with social commentary.**
- **Reflect on the theme of fate and human helplessness when faced with unexpected challenges.**
- **Develop Critical Thinking**
- **constructs discourse like news report , narration, announcement,live news reporting etc**
- **communicate effectively in the target language.**

Resources:

videos, pictures slides, CB

Process	Assessment
Introductory interaction **Teacher shows a video clip from the movie" Vellanakalude naadu". initiates a discussion** • **What do you feel about the helplessness of the owner?** • **Have you noticed such situations where human stupidity creates trouble?** • **Have you heard of the expression "put the cart before the horse"?** **Teacher asks the slow pace learners to say what they feel special about the video** **Elicits responses** **Lets read a similar story** ***Process reading*** **Teacher proceeds to the text. They are asked to read the lesson from " There came to our town.....road engine** **They keep track of the reading .Teacher supports the reading process by showing some images to**	

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convey the meaning of difficult towards. initiates a discussion.

- **Who came to the town some years ago?**
- **How did the people of the district respond to the show?**
- **Why do you think people were attracted to Gaiety Land?**
- **Which of the following was NOT mentioned as part of the side-shows?**

  a) Performing parrots

  b) Crack motorcyclists

  c) Shooting galleries

  d) Puppet shows

Elicits responses.

Discourse

Teacher asks the students to prepare and present a announcement of the show

Teacher includes slow pace learners and asks them to take part in it ,They attempt the discourse, group refinement,presentation Editing. Teacher records the presentations and suggests modifications.

Teacher proceeds to the next section. They are asked to read paragraphs from "One evening they drew the ticket number.....safe through"

They keep track of their reading. teacher asks some global questions to ensure reading. learners sit in groups and share their findings. they refer to the glossary. Teacher supports the reading process in groups. Teacher shows some slides to convey the meanings of words

initiates the discussion

- **Why was the narrator stunned when his number was declared?**
- **Why couldn't the narrator transport the road engine immediately?**
- **How did the showman respond when the narrator asked if he could sell the engine to a municipality?**
- **How did the narrator show his affection towards the road engine?**
- **What did the showman agree to do with the road engine?**

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a) Sell it for the narrator
b) Repair it before delivery
c) Keep it on the grounds till the end of the season
d) Transport it to the narrator's home

- **Who sent the narrator a notice to collect the road engine when the show ended?**
  a) His friends
  b) The showman
  c) The municipality
  d) A driver
  Elicits responses

Teacher includes the slow pace learners by asking them simple questions

Discourse

Prepare a warning letter from the Municipality to the owner

They attempt the discourse, group refinement, presentation Editing. Teacher records the presentations and suggests modifications.

Assignment

prepare a similar instruction manual for operating a mobile phone.

Teacher proceeds with the next section .They are asked to read paragraphs from "Soon I received a second .....engine driver".

They keep track of their reading. Teacher asks some global questions to ensure reading. Teacher gives some contexts and some images to get the meanings. Initiates a discussion

- **what did the second notice contain?**
- **What was the rent he paid?**
- **How did he manage to pay it?**
- **Why did he become desperate?**
- **Why didn't he get a driver?**

Elicits responses.

Discourse

The narrator tries to convince the municipal chairman to buy the road engine.

Now, let's enact the scene after preparing a script.

They attempt the discourse ,group refinement, presentation Editing .Teacher records the presentations and suggests modifications..

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Teacher proceeds with the next section .They are asked to read paragraphs from " Meanwhile the municipality .....marched me off."
They keep track of their reading. Teacher asks some global questions to ensure reading. Teacher gives some contexts and some images to get the meanings. Initiates a discussion

- **What was the help offered by the priest?**
- **What was his plan ?**
- **What kind of a driver was Joseph?**
- **How did the engine behave?**
- **How did the behaviour of the crowd affect the elephant?**
- **What did the elephant do after trumpeting?**
  a) Sat down quietly
  b) Strained and snapped its ropes
  c) Walked away calmly
  d) Ran around in circles

Elicits responses.

<u>Discourse</u>

Write a news article, reporting the chaos that erupted when the narrator attempted to move the road engine.
They attempt the discourse ,group refinement presentation Editing .Teacher records the presentations and suggests modifications..

<u>Assessment</u>

Write the engine owner's thoughts about the incident
Teacher proceeds to the last section .They are asked to read paragraphs from "When I was released from .....refused to accept"
They keep track of their reading. teacher asks some global questions to ensure reading. Teacher gives some contexts and some images to get the meanings. Initiates a discussion

- **"I was at a loss" Why did he say so?**
- **How did Nature came to his help?**
- **What did he see when he went t see the engine?**
- **Why was the owner of the well happy?**

Elicits response

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Discourse **Imagine you are a reporter, Prepare a Live News Reporting of the earthquake which threw the engine to the well** **They attempt the discourse ,group refinement presentation Editing .Teacher records the presentations and suggests modifications..**	

TEACHING MANUAL 3

STD 8 The Trail of Science **Date:**

Name of the teacher:

Lesson: The Astronomer

prds. 3

Theme:The conflict between scientific knowledge and human emotion

Sub themes:Innocence and imagination of childhood

Learning outcomes:

- **Identify the contrast between scientific inquiry and emotional sensitivity.**
- **Analyse the literary devices used in the poem.**
- **enhance creative thinking.**
- **Read aloud with proper intonation and rhythm**
- **Learn to analyse a poem critically and connect the theme with life**

Resources:

videos, pictures slides, CB

Process	**assessment**
Introductory Interaction **with the help of AI Teacher shows how a picture is created using apt prompts. Asks them to create some prompts for an image in which a boy is seen looking at the moon. Initiates a discussion** ● **How does the boy look while gazing at the moon? Curious? Happy? Thoughtful?** ● **What do you think the boy is thinking about while looking at the moon?** ● **Have you ever watched the moon at night? How did you feel?** **Elicits responses** **Here is a beautiful poem by Tagore**	

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Process reading **Teacher recites the poem emphasizing the emotions Asks them to read silently and visualize the poem Interacts with them and clears the doubts in groups** **Initiates a short discussion** ➔ **What do you think of the child's question in the first stanza? Have you ever felt the same way?** ➔ **What is Dada's answer to the child's question?** ➔ **Do you think the child is convinced by Dada's arguments? Why/Why not?** ➔ **How does the poem capture the innocence and imagination of childhood?** ➔ **What does the conversation between the child and Dada reveal about their relationship?** ➔ **Why does the child compare the moon to the mother's face? Do you think the child's logic is mistaken, or truly insightful?** ➔ **What is the difference in perception between the child and Dada? Which perspective appears more convincing to you? Why?** **Teacher proceeds to the analysis of the poem** **Divides the class into groups and asks them to present imagery ,contrast etc. They are asked to write an analysis of it. They attempt the discourse Individual attempt,group refinement,** **presentation Editing,Then a teacher's version is given** **Assessment** **page 93 activity B**	